

Dual of expression

- Dual of Boolean expression:

Change the following in the expression to find the dual.

0		1
1		0
+		.
.		+

Complement of expression

- Complement of Boolean expression:

Change the following in the expression to find the complement.

0	→	1
1	→	0
+	→	.
.	→	+
$\bar{x}$	→	x
x	→	$\bar{x}$

Note:

Complement of Boolean expression can be found by applying DeMorgan's Theorem.

Example:

$$Q: F = A'C + A'B + AB'C + BC$$

For the above Boolean function

- (a) Find the dual of expression.
- (b) Find the complement of expression.

Ans:

a) Dual of $F = (A' + C).(A' + B).(A + B' + C).(B + C)$

b) Complement of $F = F' = (A + C').(A + B').(A' + B + C').(B' + C')$

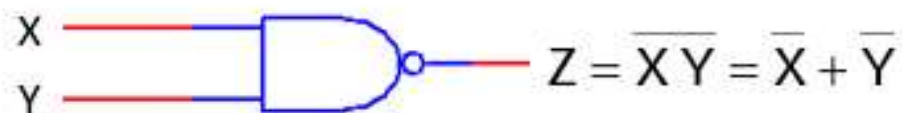
➤ Advantage of minimization of logical expression:

- Number of logic gates will decrease.
- Speed of operation will increase
- Power dissipation will decrease
- Cost will decrease
- Complexity of circuit will decrease
- Fan-in may be reduced.

Universal Gates: NAND and NOR Gates

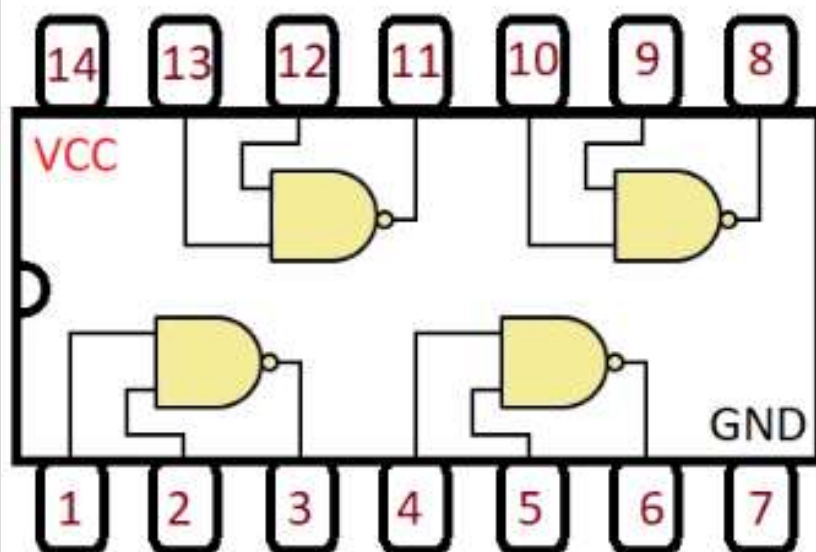
- A universal gate is a gate which can implement any Boolean function without need to use any other gate type.
- The NAND and NOR gates are universal gates.
- Implementing the logic circuits using NAND and NOR gates are economical and easier to fabricate and are the basic gates used in all IC digital logic families.
- Implementation of a Boolean function using NAND gates require the function to be in SOP form.
- Implementation of a Boolean function using NOR gates require the function to be in POS form.

NAND Gate

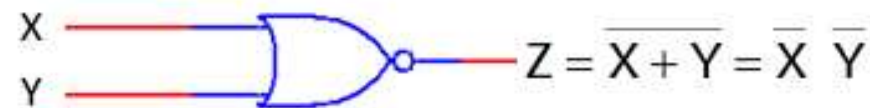


X	Y	Z
0	0	1
0	1	1
1	0	1
1	1	0

IC 7400

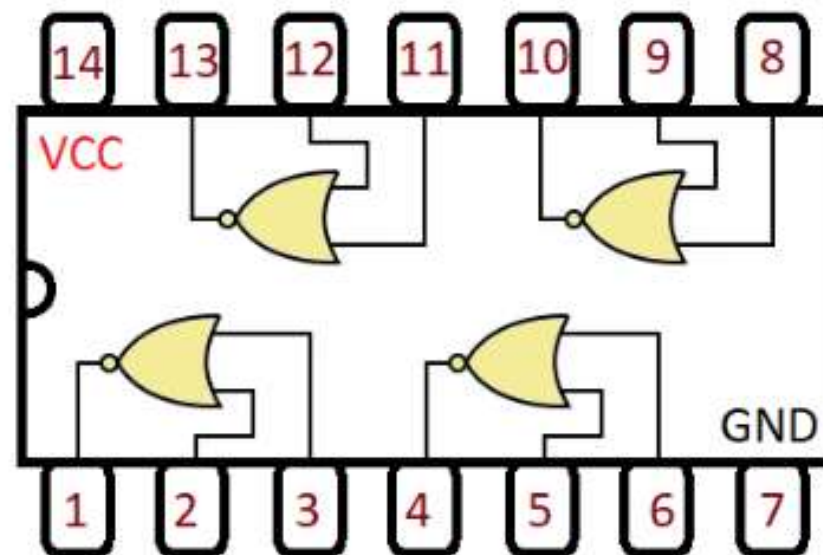


NOR Gate



X	Y	Z
0	0	1
0	1	0
1	0	0
1	1	0

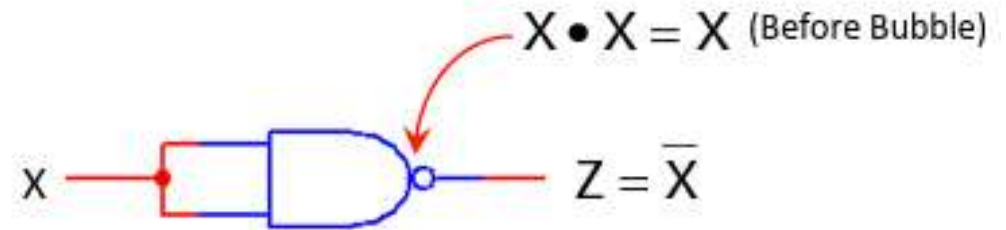
IC 7402



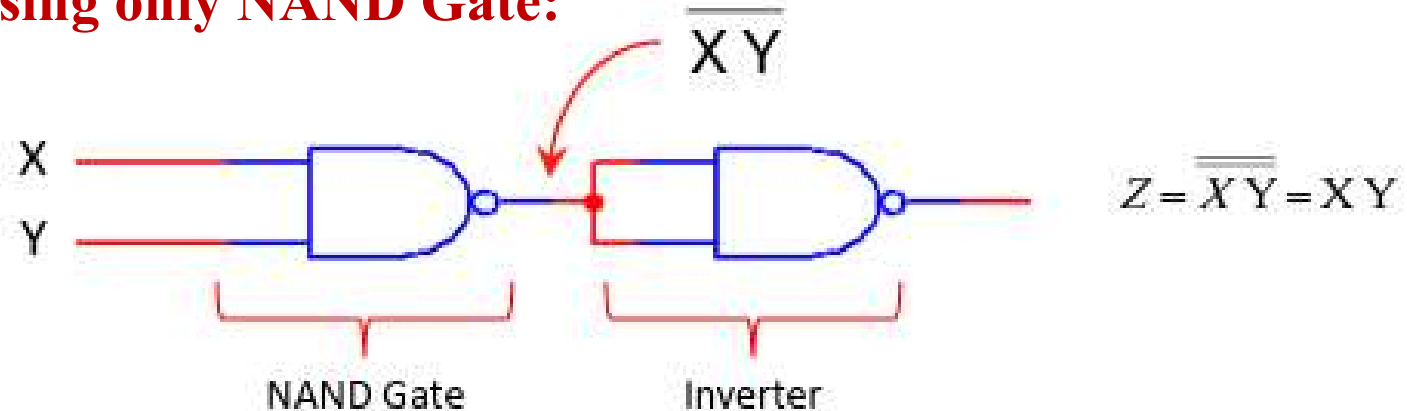
➤ Inverter Using only NAND Gate:

X	Z
0	1
1	0

} Equivalent to Inverter



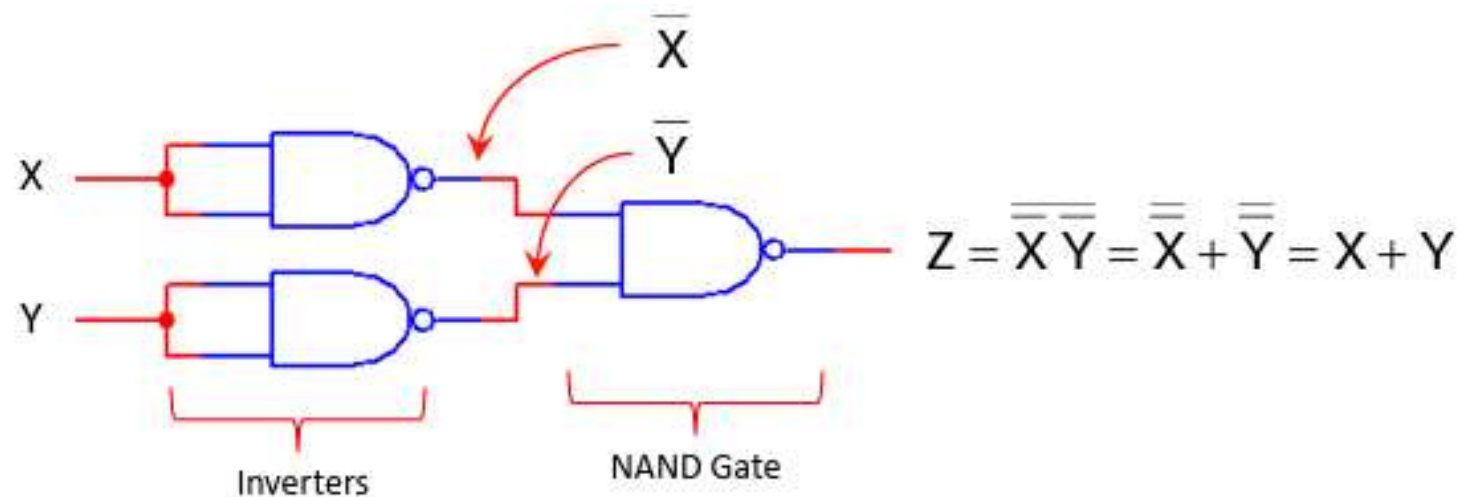
➤ AND Gate Using only NAND Gate:



X	Y	Z
0	0	0
0	1	0
1	0	0
1	1	1

} Equivalent to AND Gate

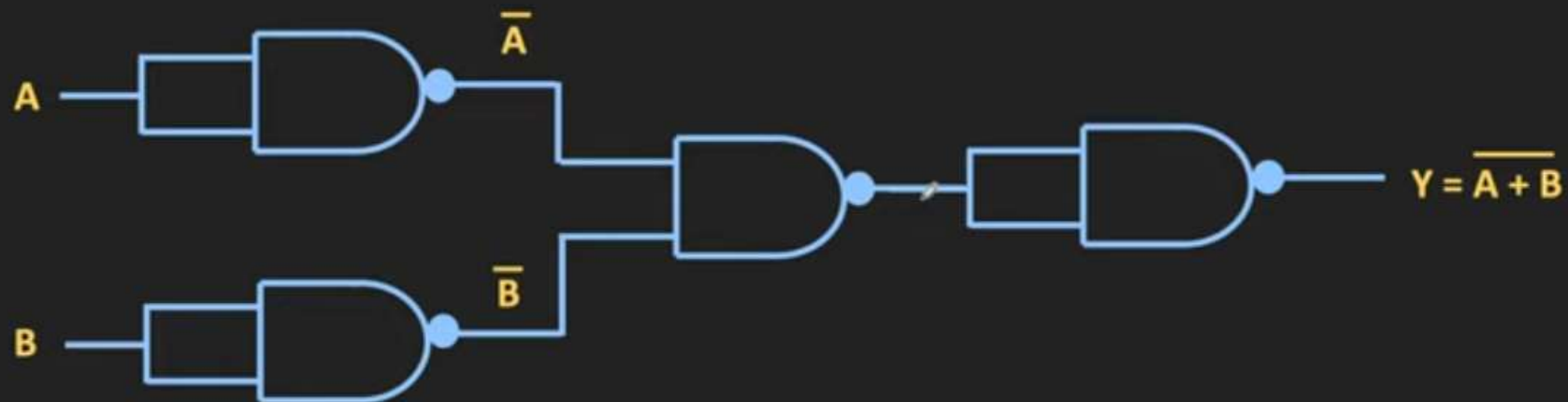
➤ OR Gate Using only NAND Gate:



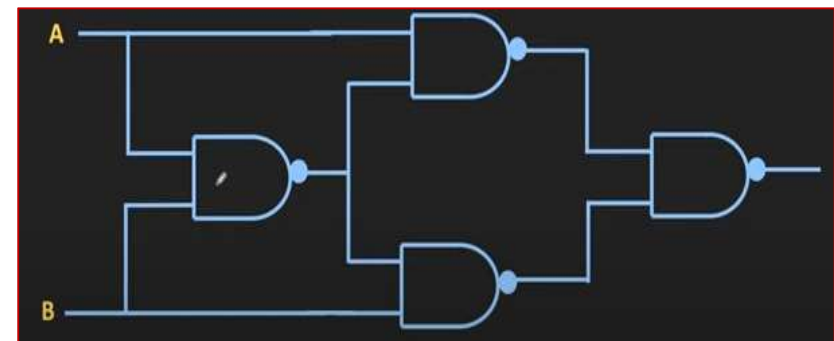
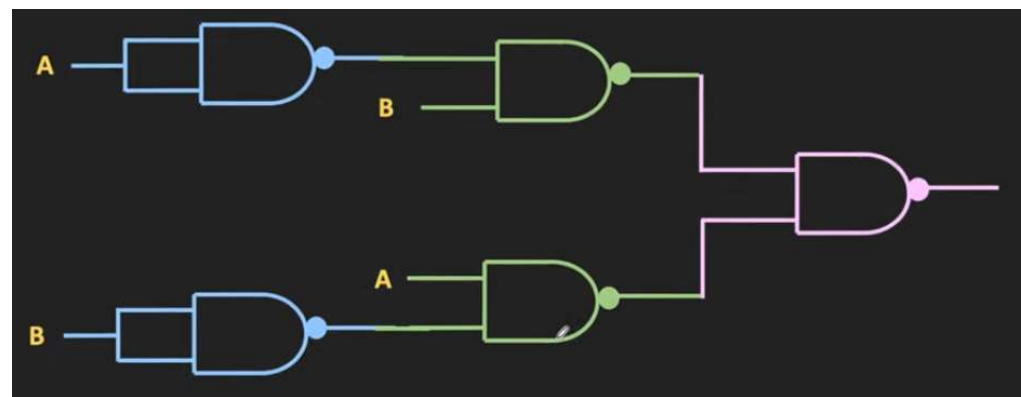
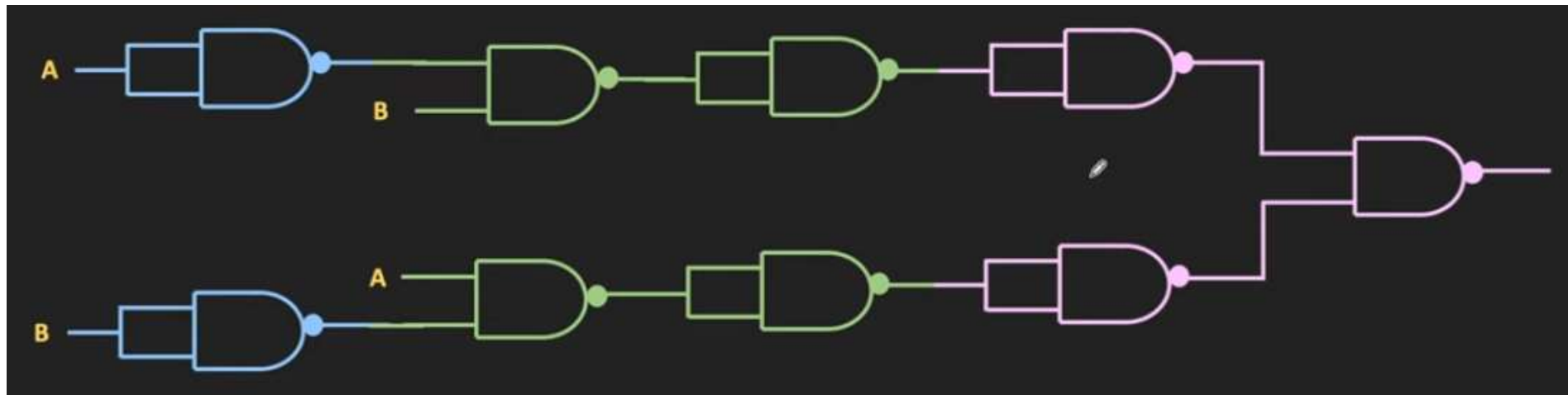
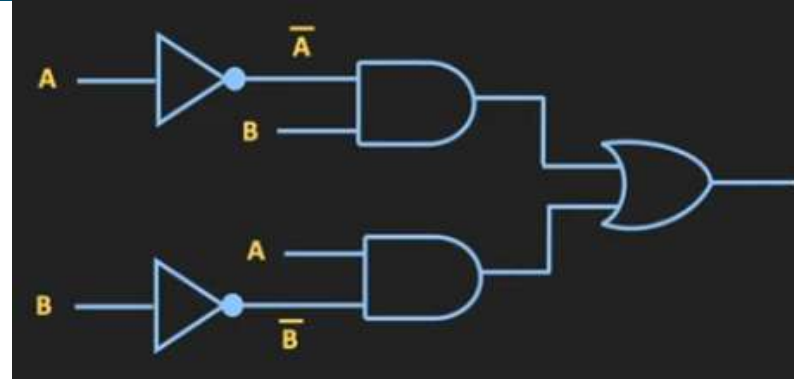
X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	1

Equivalent to OR Gate

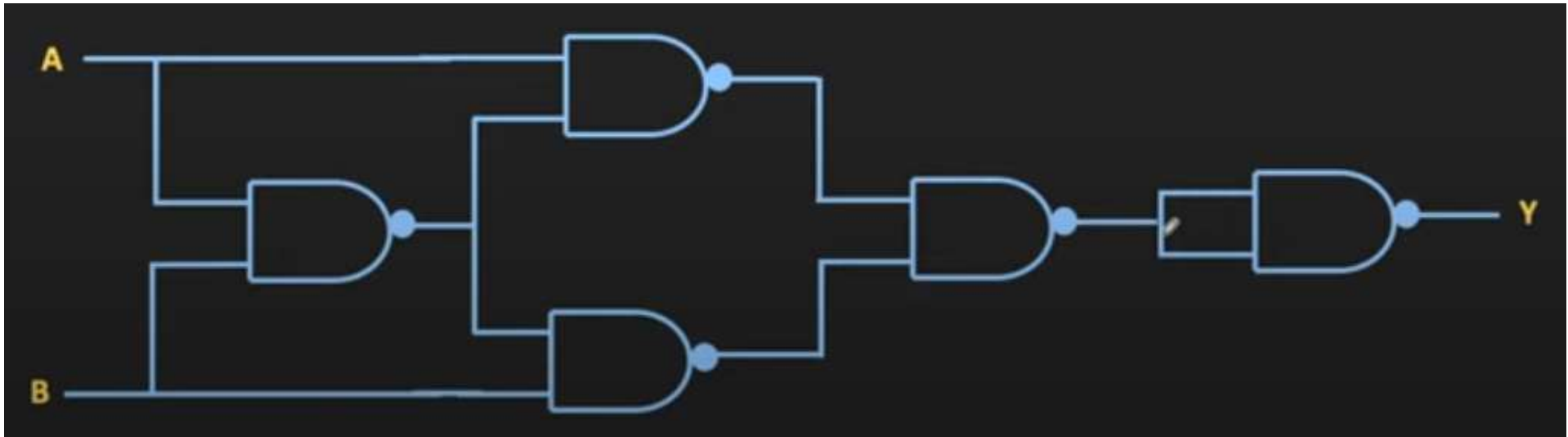
NOR Gate using NAND Gates



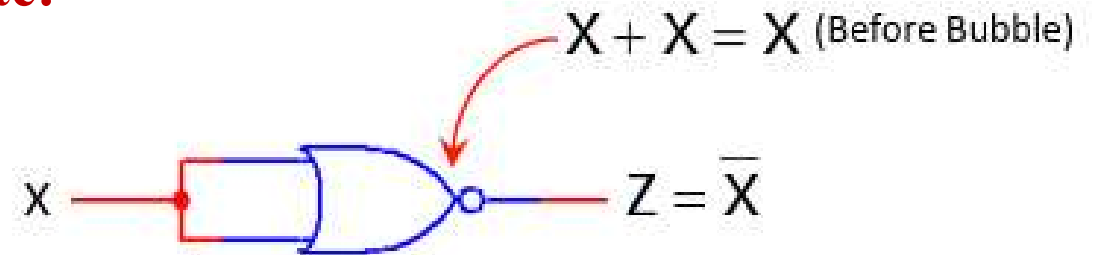
XOR Gate using NAND Gates



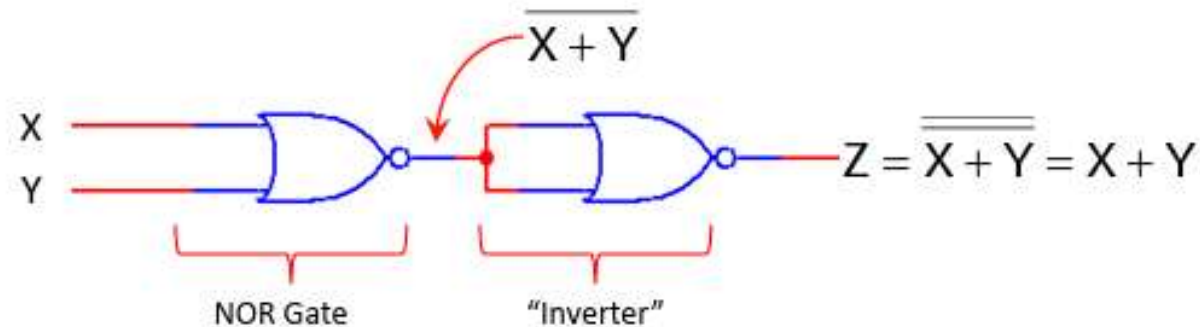
XNOR Gate using NAND Gates



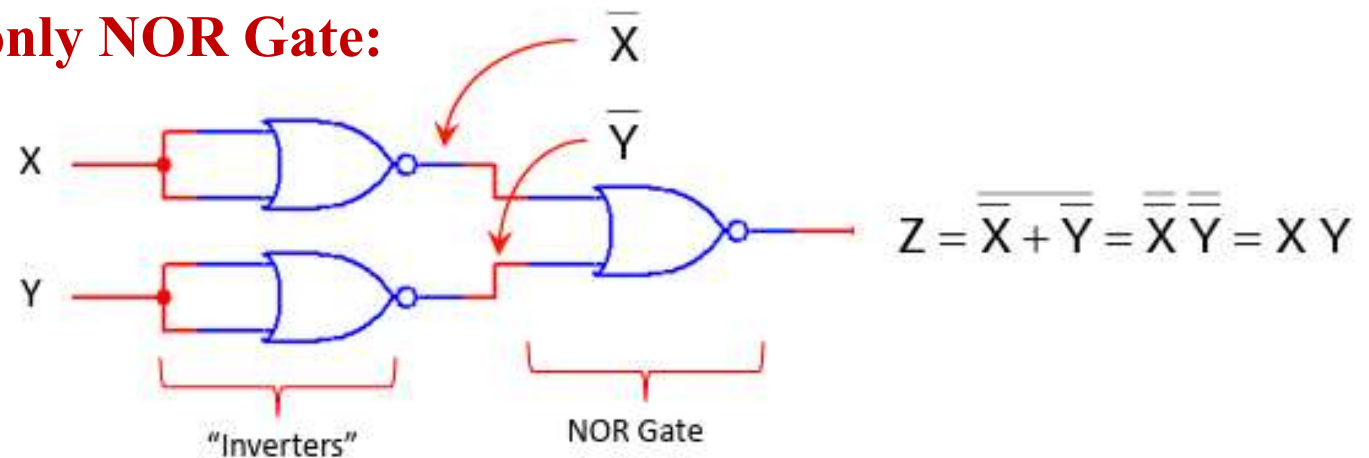
➤ Inverter Using only NOR Gate:



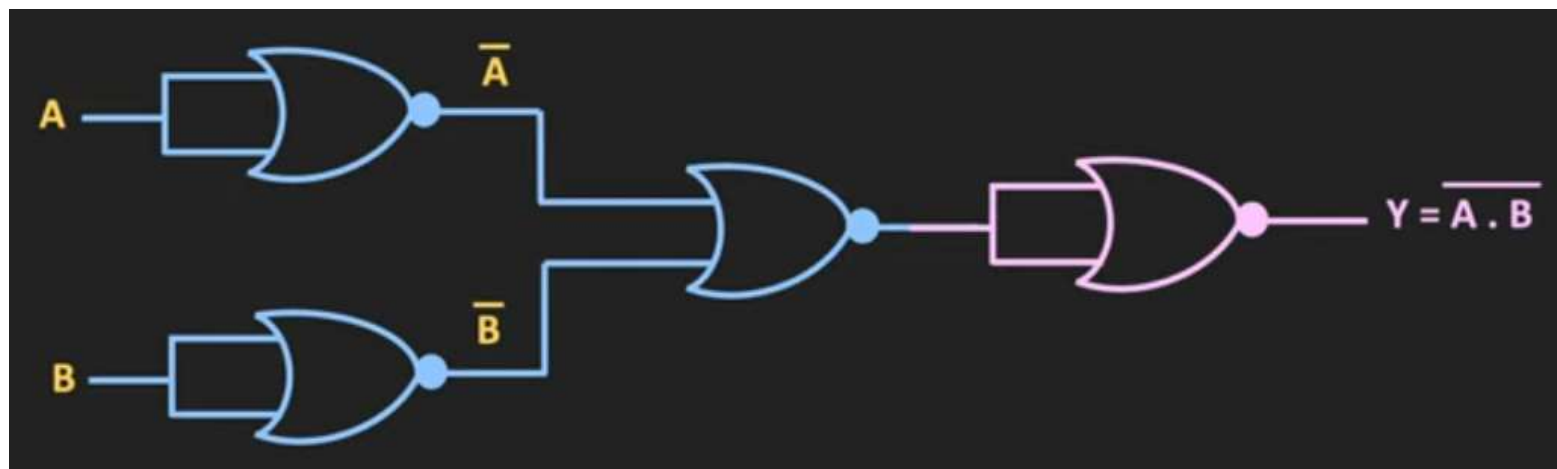
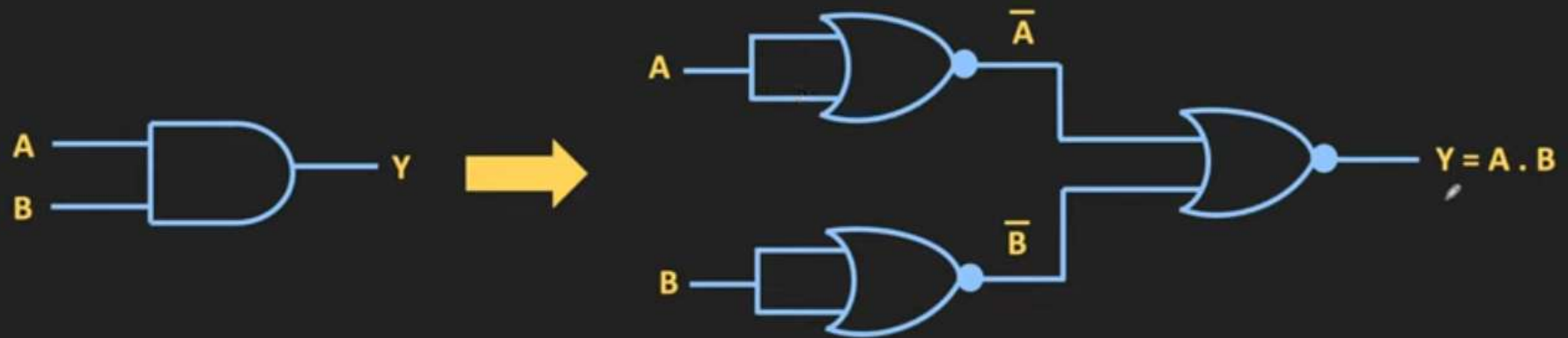
➤ OR Gate Using only NOR Gate:



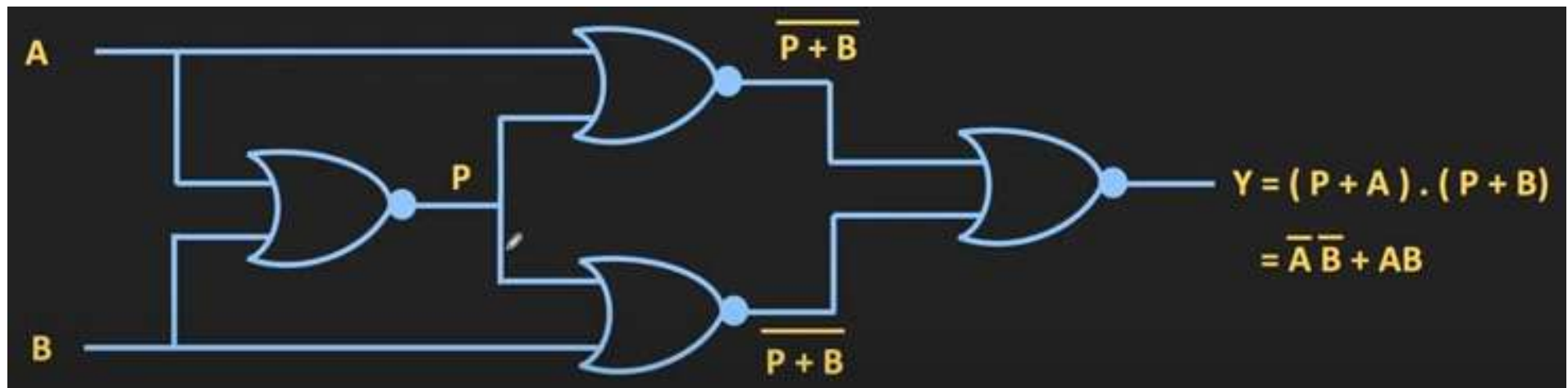
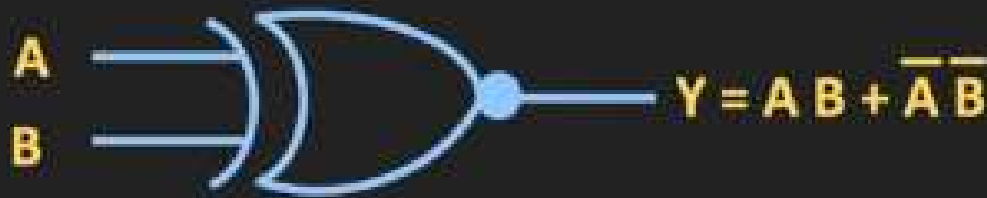
➤ AND Gate Using only NOR Gate:



NAND Gate using NOR gate



XNOR Gate using NOR Gates

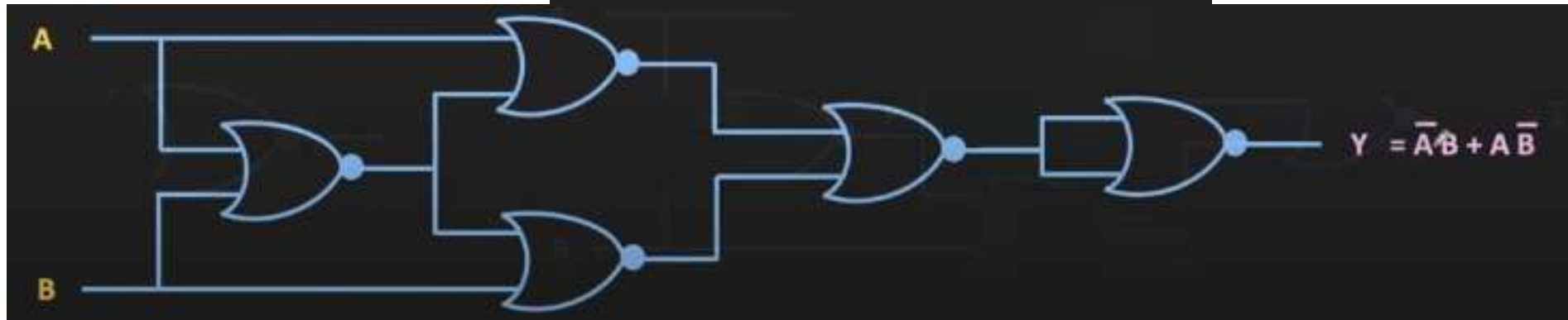


$$A \oplus B = \overline{A}B + A\overline{B}$$

$$A \odot B = AB + \overline{A}\overline{B}$$

$$\overline{A \odot B} = A \oplus B$$

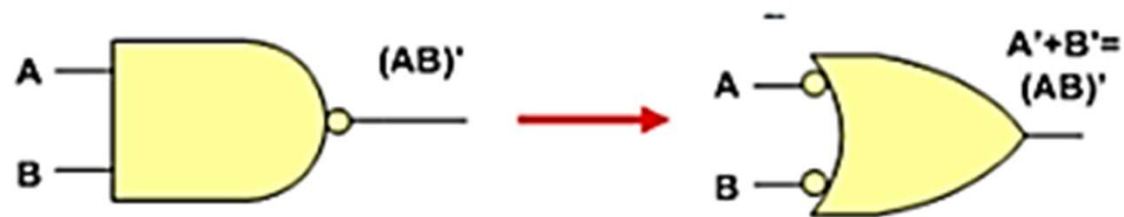
$$\overline{AB + \overline{A}\overline{B}} = A\overline{B} + \overline{A}B$$

XOR Gate using NOR Gates

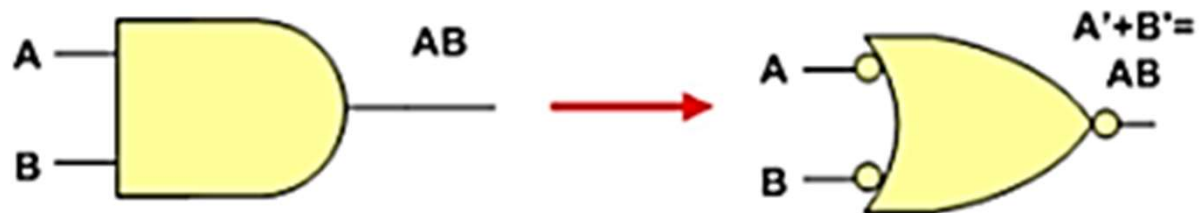
Logic Gates	No. of NAND Gate	No. of NOR Gate
NOT	1	1
AND	2	3
OR	3	2
XOR	4	5
XNOR	5	4
NOR	4	1
NAND	1	4

Gate Equivalence

A NAND Gate is equivalent to an inverted-input OR gate

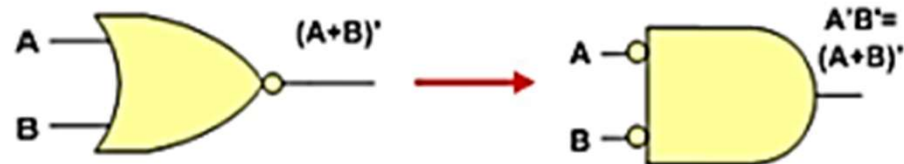


An AND gate is equivalent to an inverted-input NOR gate

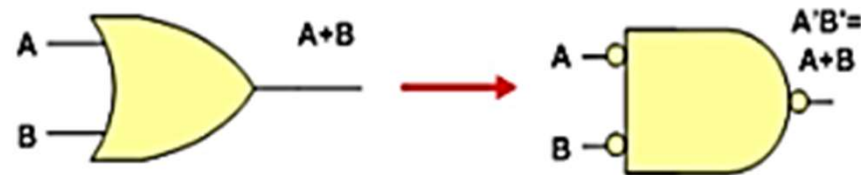


Gate Equivalence

A NOR gate is equivalent to an inverted-input AND gate.



An OR gate is equivalent to an inverted-input NAND gate.



Two NOT gates in series are same as a buffer because they cancel each other

as $A'' = A$



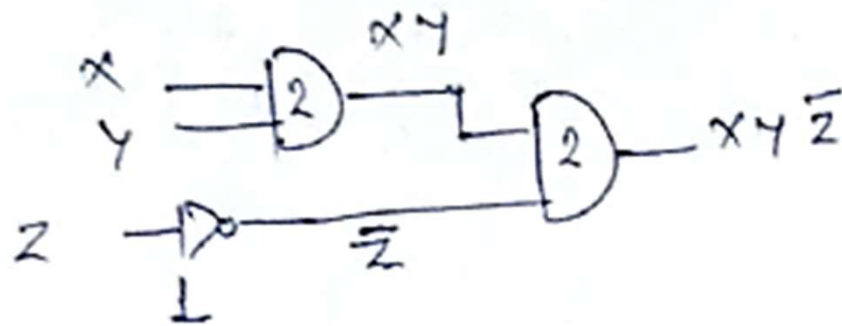
➤ Procedure for obtaining NAND only circuits:

- Simplify the function and express it in SOP form
- First stage - Draw a NAND gate for each product term of the expression that has at least 2 literals. The inputs to each NAND gate are the literals of the term.
- Second stage – Draw a single NAND gate(or invert OR). The first stage outputs will be the inputs of the second stage NAND gate.
- Any term in the expression with single literal requires an inverter in the first stage.

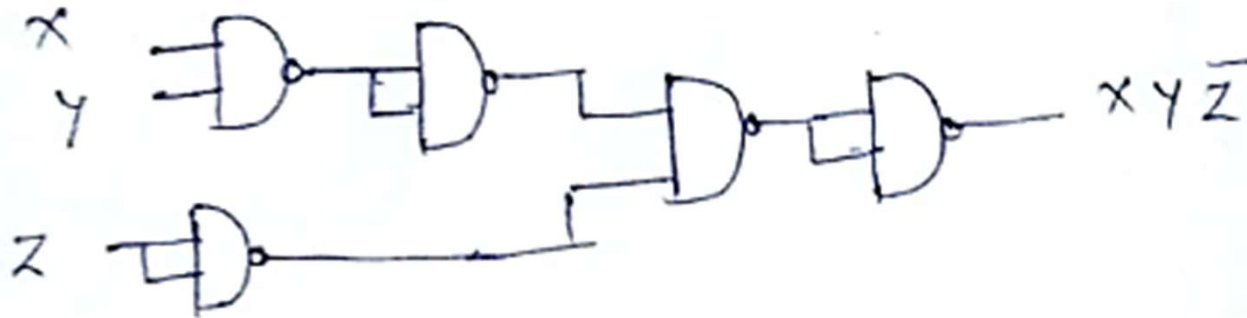
➤ Procedure for obtaining NOR only circuits:

- Simplify the function and express it in POS form
- First stage - Draw a NOR gate for each sum term of the expression that has at least 2 literals. The inputs to each NOR gate are the literals of the term.
- Second stage – Draw a single NOR gate(or invert AND). The first stage outputs will be the inputs of the second stage NOR gate.
- Any term in the expression with single literal requires an inverter in the first stage.

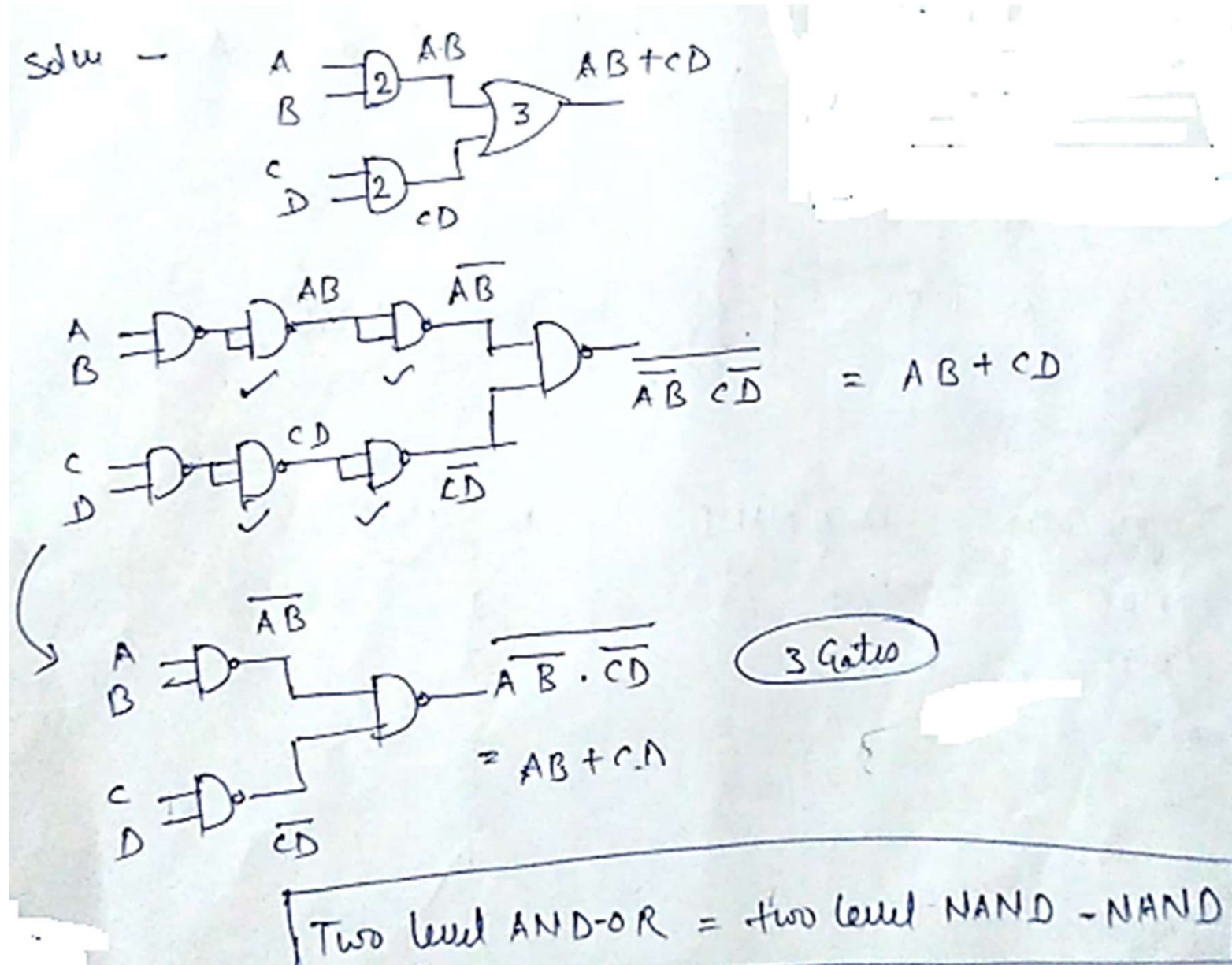
Q. Implement $XY\bar{Z}$ using two input NAND gate.



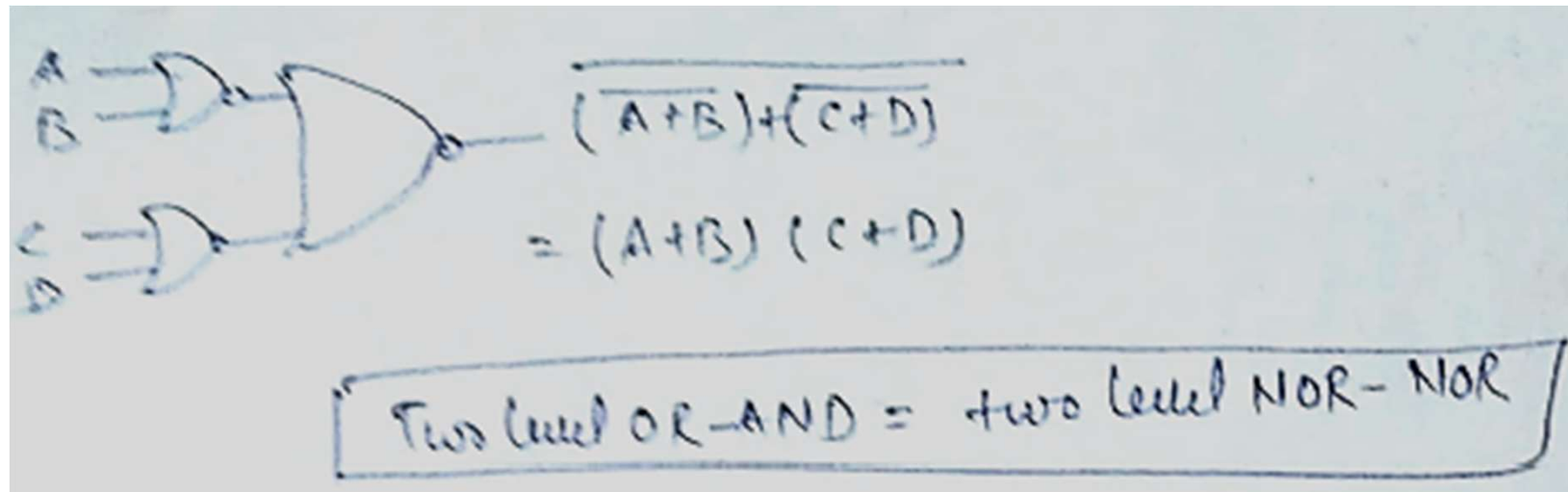
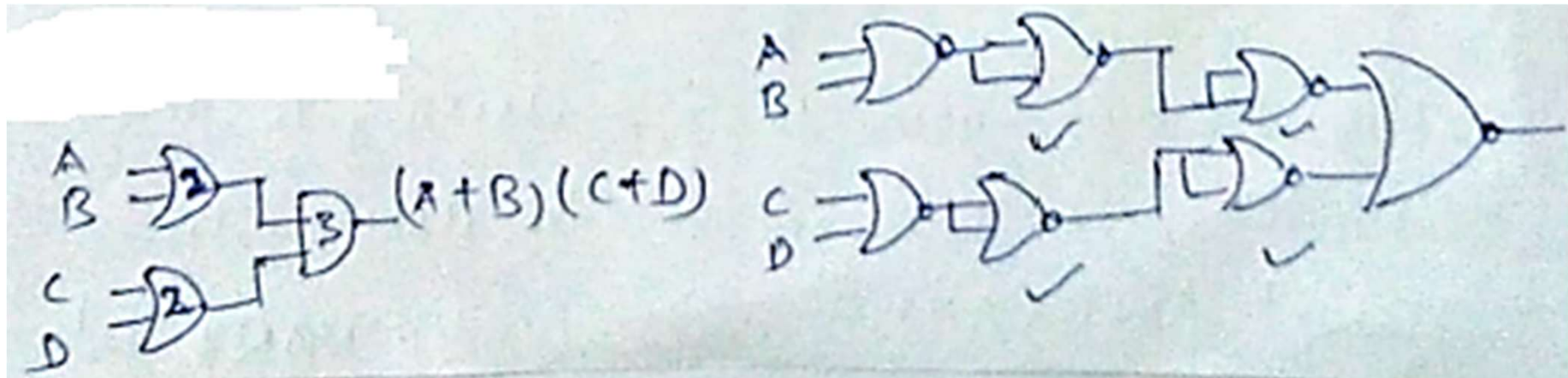
for NOT one NAND
for AND two NAND



Q. Implement $AB + CD$ using two input NAND gate.

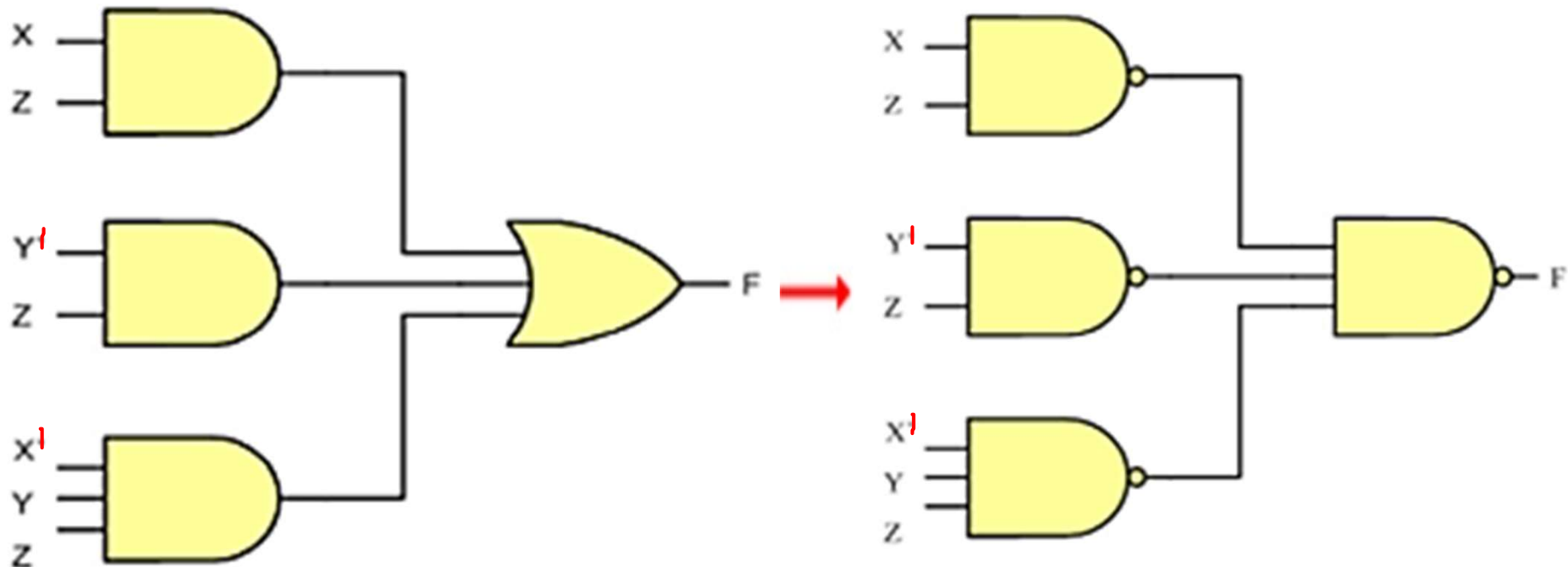


Q3. Implement $(A + B)(C + D)$ using two input NOR gate.



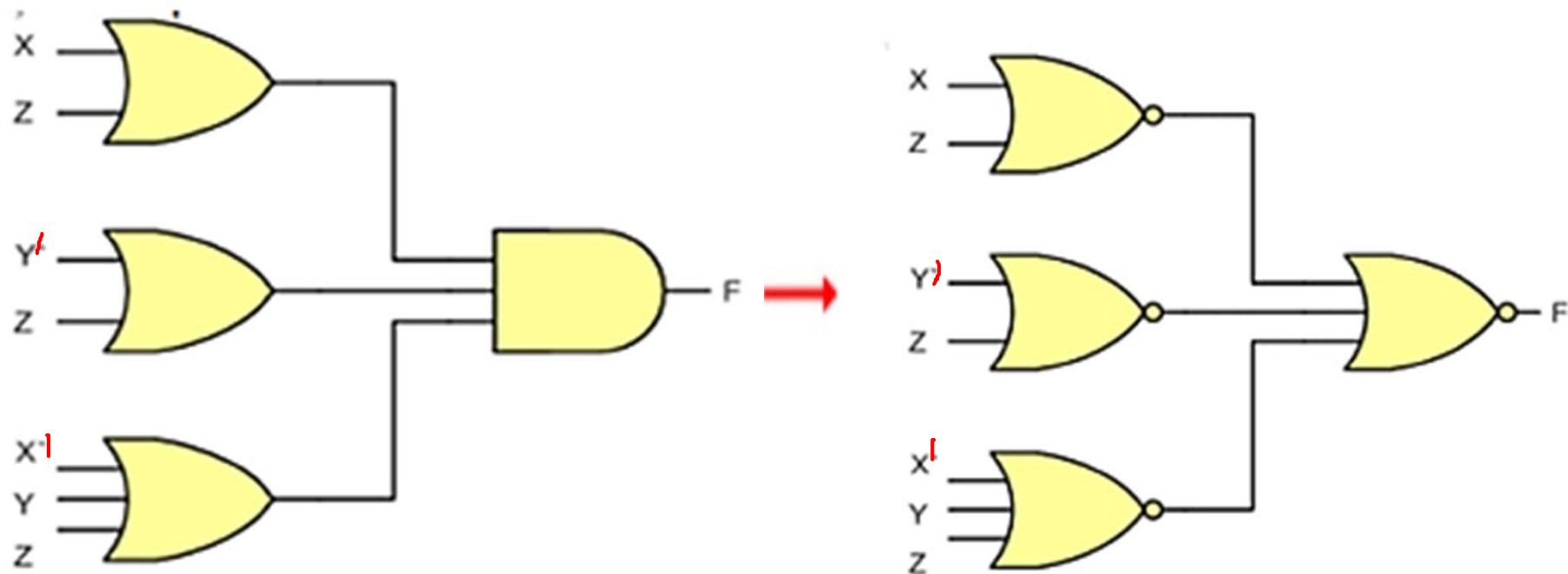
Implement the SOP function $F = XZ + Y'Z + X'YZ$ using NAND gates only

Soln: Being an SOP expression, it is implemented in 2-levels as shown below.

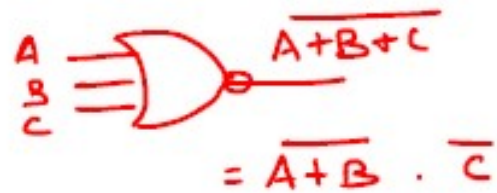


Implement the POS function $F = (X+Z)(Y'+Z)(X'+Y+Z)$ using NOR gates only

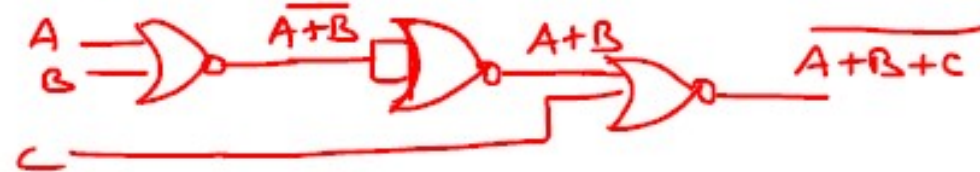
Soln: Being an POS expression, it is implemented in 2-levels as shown below.



3 i/p NOR gate using 2 i/p NOR gates

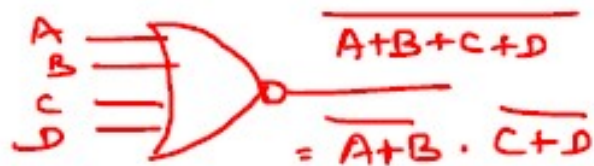


$\equiv$

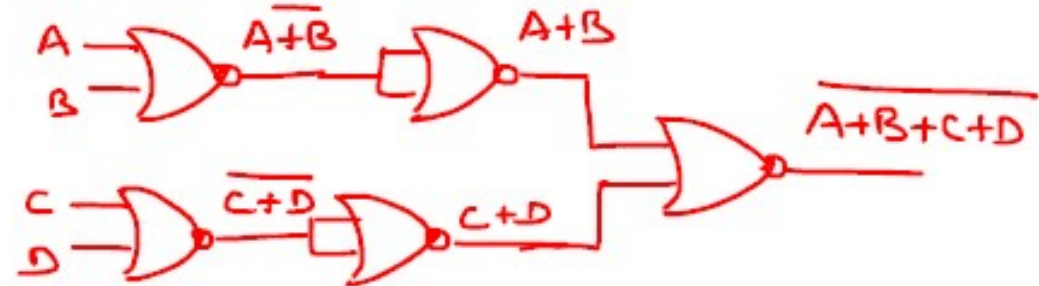


and $\equiv$ inv NOR.

4 i/p NOR using 2 i/p NOR gates



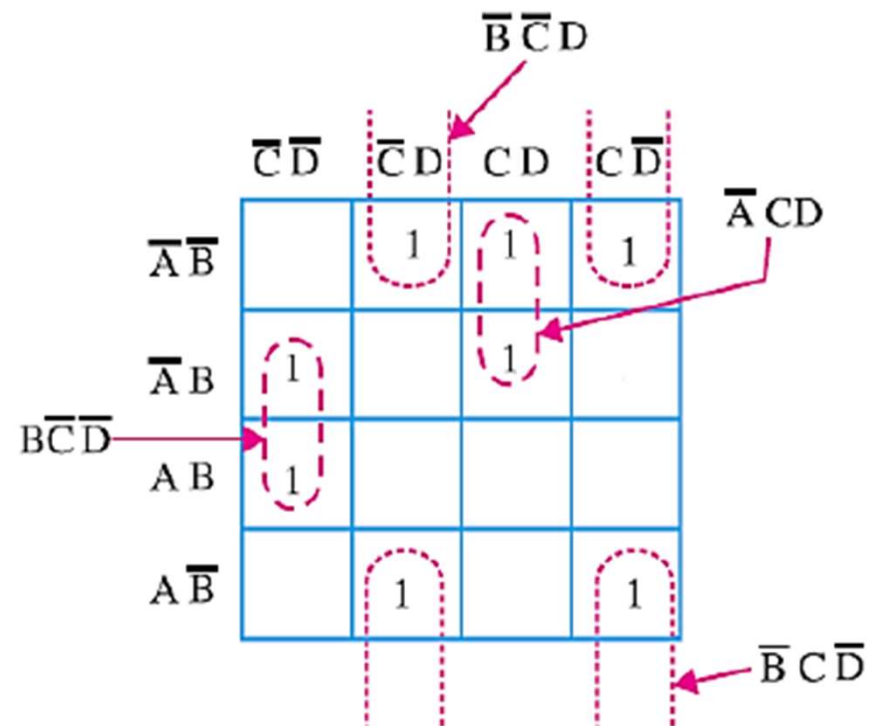
$\equiv$



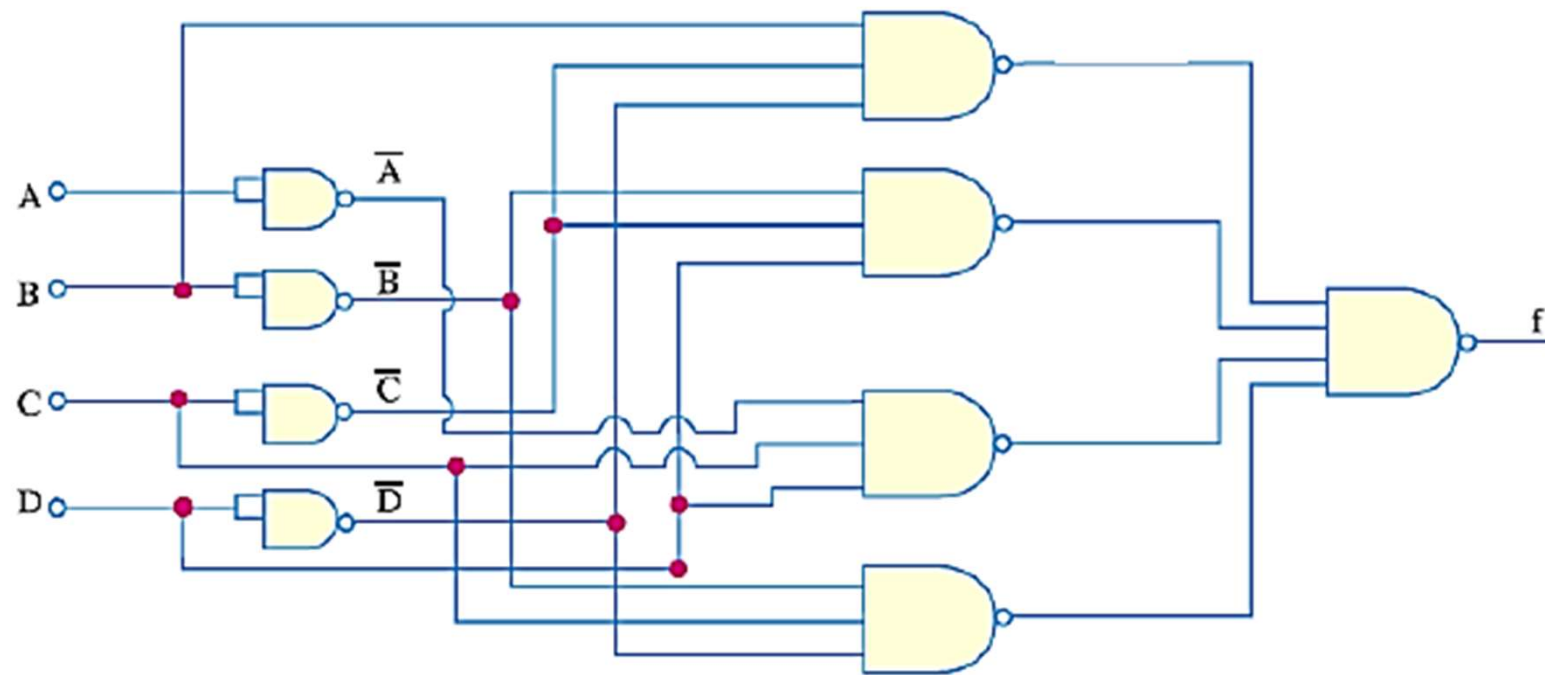
Implement the following Boolean expression using minimum number of 3-input NAND gates.

$$f(A, B, C, D) = \Sigma(1, 2, 3, 4, 7, 9, 10, 12)$$

Decimal Number	Inputs				Output f
	A	B	C	D	
0	0	0	0	0	0
1	0	0	0	1	1
2	0	0	1	0	1
3	0	0	1	1	1
4	0	1	0	0	1
5	0	1	0	1	0
6	0	1	1	0	0
7	0	1	1	1	1
8	1	0	0	0	0
9	1	0	0	1	1
10	1	0	1	0	1
11	1	0	1	1	0
12	1	1	0	0	1
13	1	1	0	1	0
14	1	1	1	0	0
15	1	1	1	1	0



$$f(A, B, C, D) = B\bar{C}\bar{D} + \bar{B}\bar{C}D + \bar{A}CD + \bar{B}C\bar{D}$$



$$f(A, B, C, D) = \overline{B}\overline{C}\overline{D} + \overline{B}\overline{C}D + \overline{A}CD + B\overline{C}\overline{D}$$

Exercise: Reduce the expression $F = \Sigma m(0,1,2,3,6,7,13,15)$ by mapping and implement in **NAND** logic.

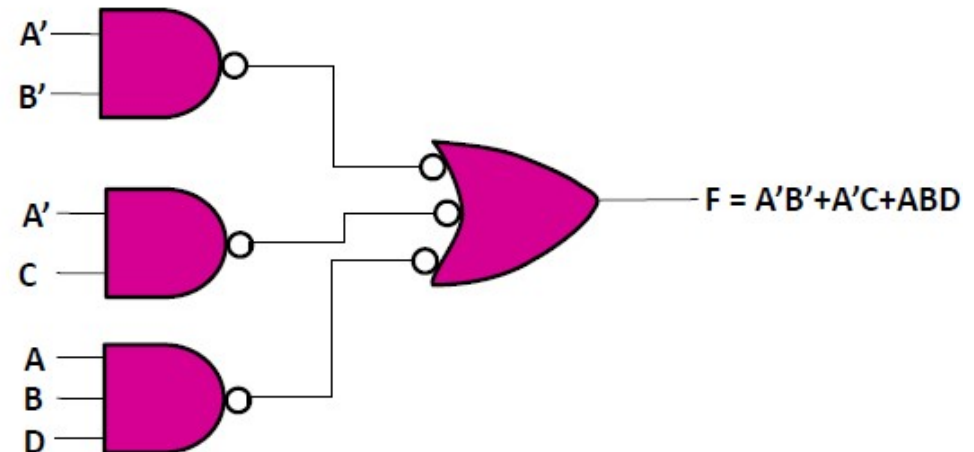
CD \ AB	00	01	11	10
00	1 ₀	1 ₁	1 ₃	1 ₂
01	0 ₄	0 ₅	1 ₇	1 ₆
11	0 ₁₂	1 ₁₃	1 ₁₅	0 ₁₄
10	0 ₈	0 ₉	0 ₁₁	0 ₁₀

$$\text{Group I} = A'B'C'D' + A'B'C'D + A'B'CD + A'B'CD' \\ = A'B'$$

$$\text{Group II} = A'BCD + A'BCD' + A'B'CD + A'B'CD' \\ = A'C$$

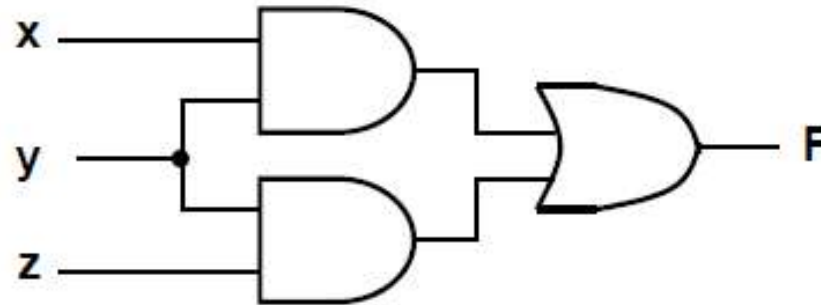
$$\text{Group III} = ABC'D + ABCD \\ = ABD$$

$$F = A'B' + A'C + ABD$$



Solve it yourself (Exercise 4.3)

- Q1. Draw a logic circuit for $BD + BE + D'F$ using only NAND gates. Use both DeMorgan method and SOP method.
- Q2. Transform the following AND-OR Circuit to NAND circuit.



- Q3. Using only NOR gates, draw a logic circuit using POS method for $(A+B+C')(B'+C'+D)$